CataList

BY CHEMISFREE

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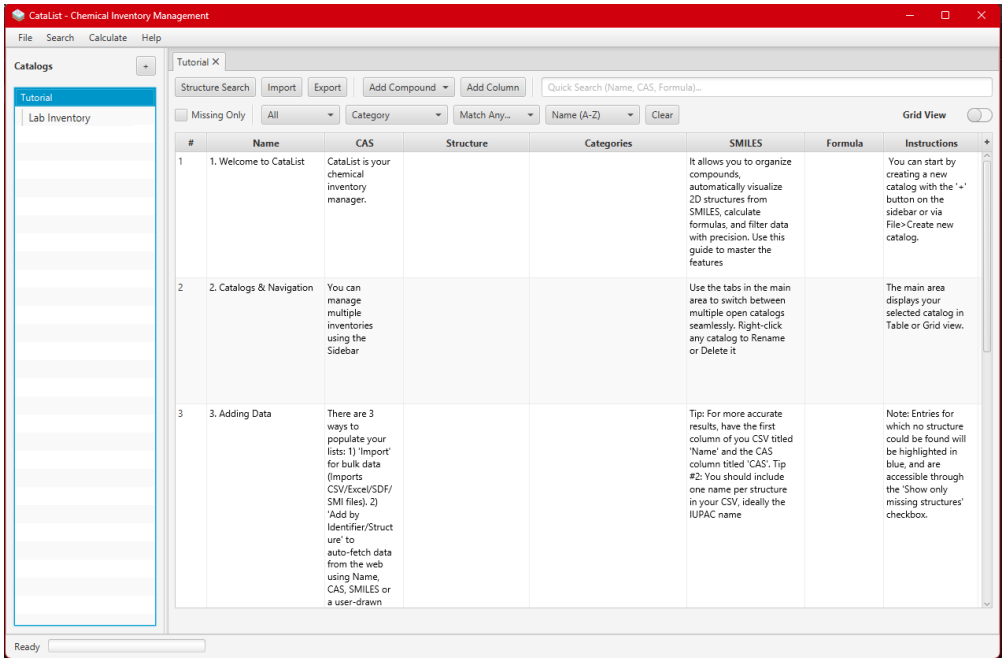
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CataList is a desktop application designed for laboratories, researchers, and stockroom managers to efficiently organize, manage, and visualize chemical inventories.

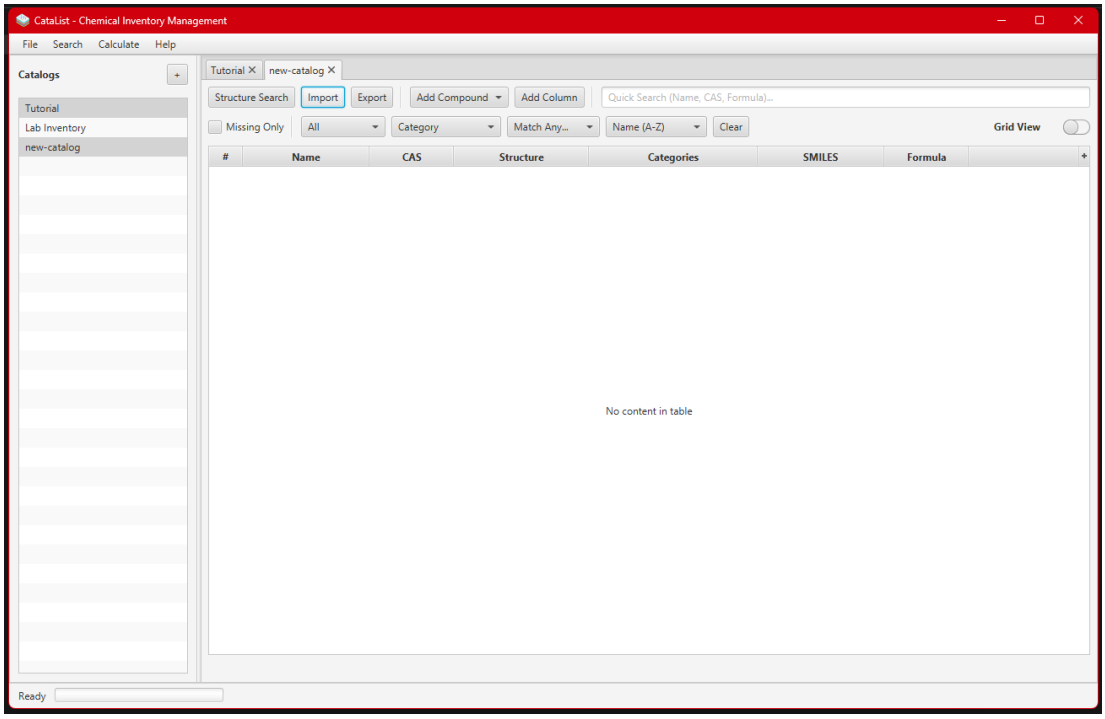
It automatically converts standard chemical identifiers into structural representations, calculates key molecular properties, and ensures data consistency across your catalog.

Getting started:

After launching **CataList**, the main interface will appear as shown below.

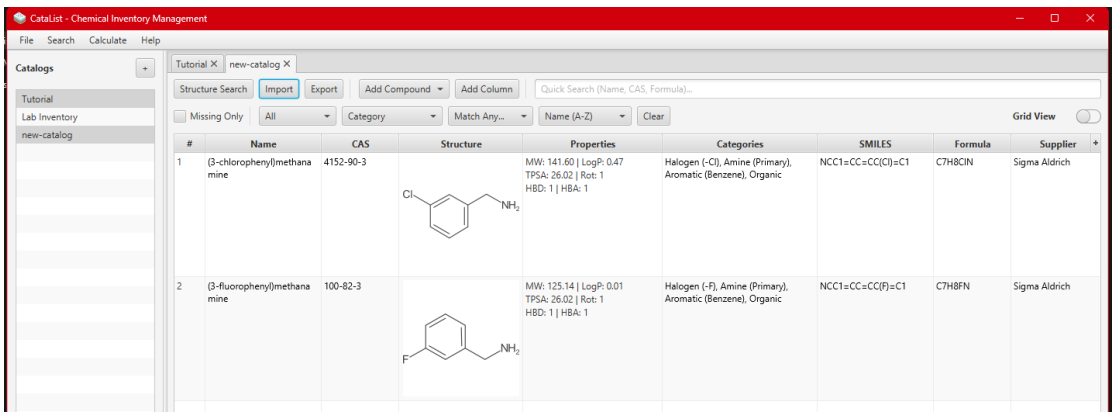


Click the “+” button to create a new catalog.
Click “Import” to load an existing catalog from a .csv, .xlsx, .sdf, or .smi file.



When a new catalog is created or imported, CataList automatically generates chemical structures from: Name or CAS number (for .csv and .xlsx files) or embedded structural information (for .sdf and .smi files).

In addition, the application computes molecular properties, generates SMILES strings, and assigns chemical categories to each compound. Any additional data present in the original file are preserved and displayed within the catalog.



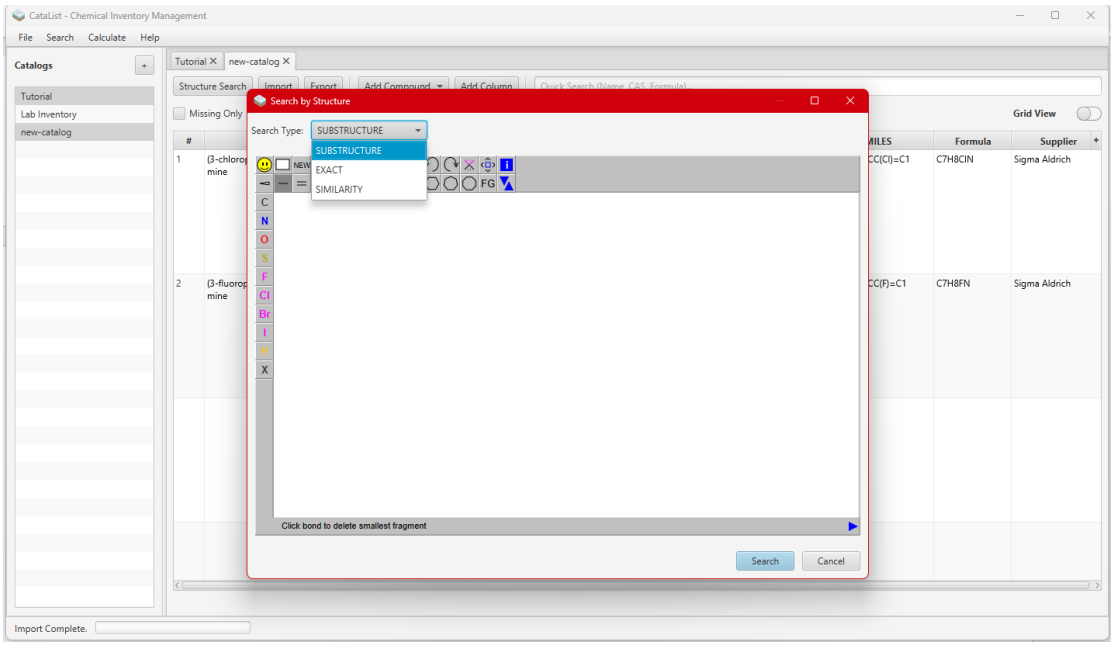
Searching and Filtering
CataList enables efficient navigation of large chemical inventories through multiple search methods:

- Category-based filtering (e.g., amines, carboxylic acids, halogens)
- Text-based search (e.g., supplier, comments, specific properties)

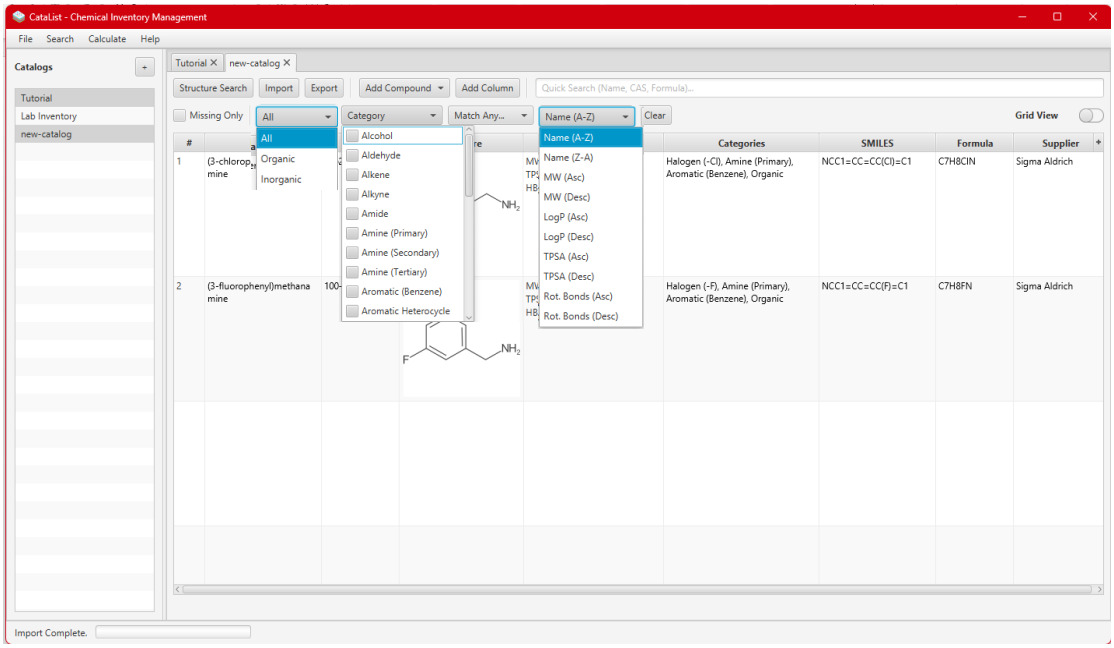
- Structure-based search via an integrated drawing tool

Structure searches support:

- Exact match
- Substructure search
- Similarity search

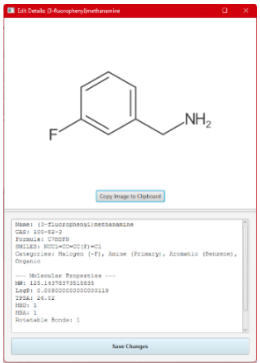


You can further refine results using the dedicated filter and sort controls.

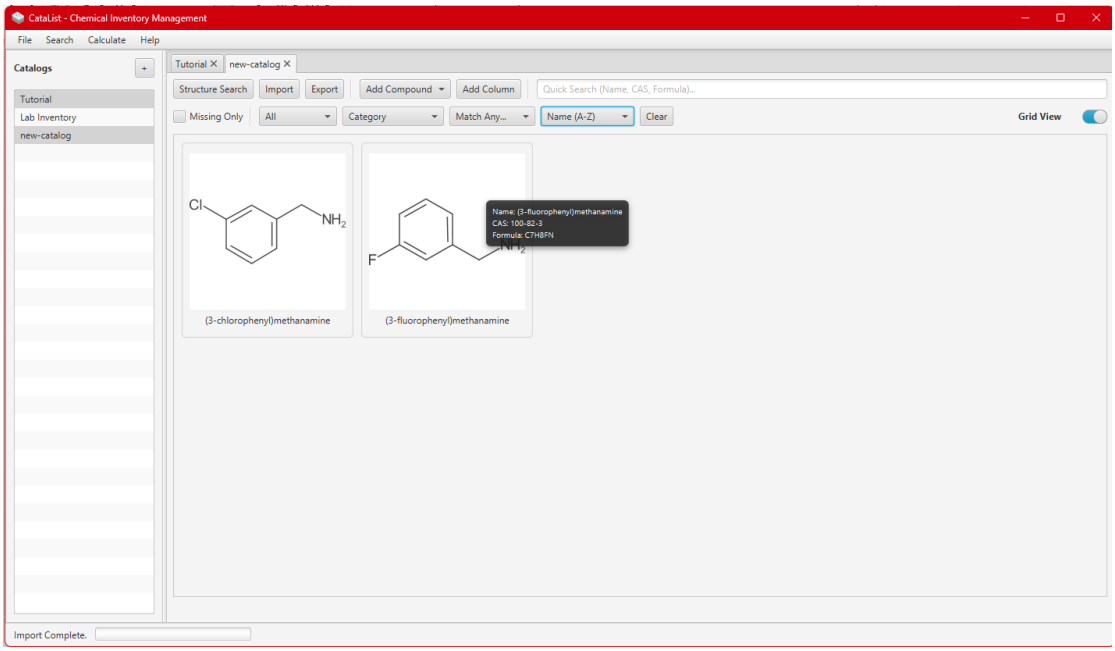


Grid View and Table View

Using the “Grid View” toggle button, you can switch between the standard table layout and a structure-based grid view. When in Grid View, hovering over a structure displays its name, CAS number, and molecular formula.

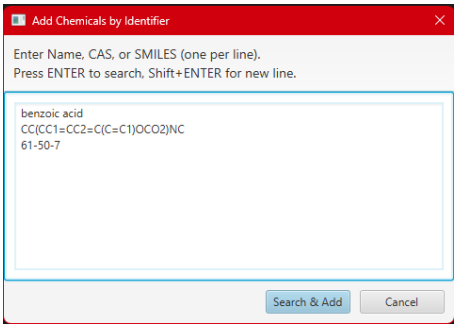


Double-clicking a structure opens a pop-up window containing an enlarged image along with the compound's full details presented in list format. From this window, you can copy the structure image, as well as copy or edit the compound's information.



Adding compounds

To add a new compound, click "Add Compound." You can then choose one of the following methods:



1. Draw Structure

- Draw the molecule using the editor.
- Click "Search."

CataList converts the drawing into a SMILES string and attempts to retrieve the corresponding Name and CAS number.

2. Add by Identifier

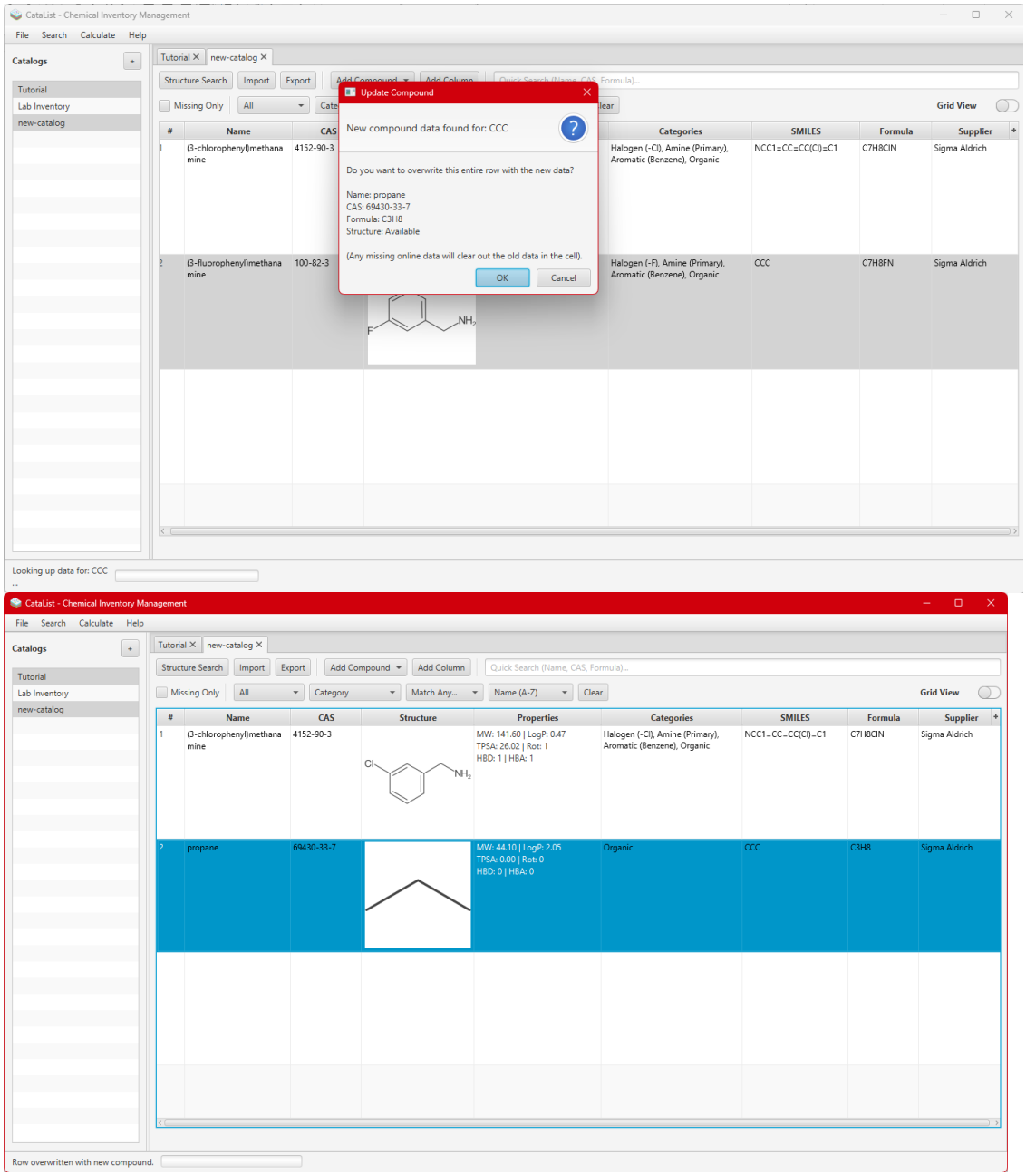
Enter one of the following:

- IUPAC name
- SMILES string
- CAS number

Batch entry is supported by entering one compound per line.

Editing Entries

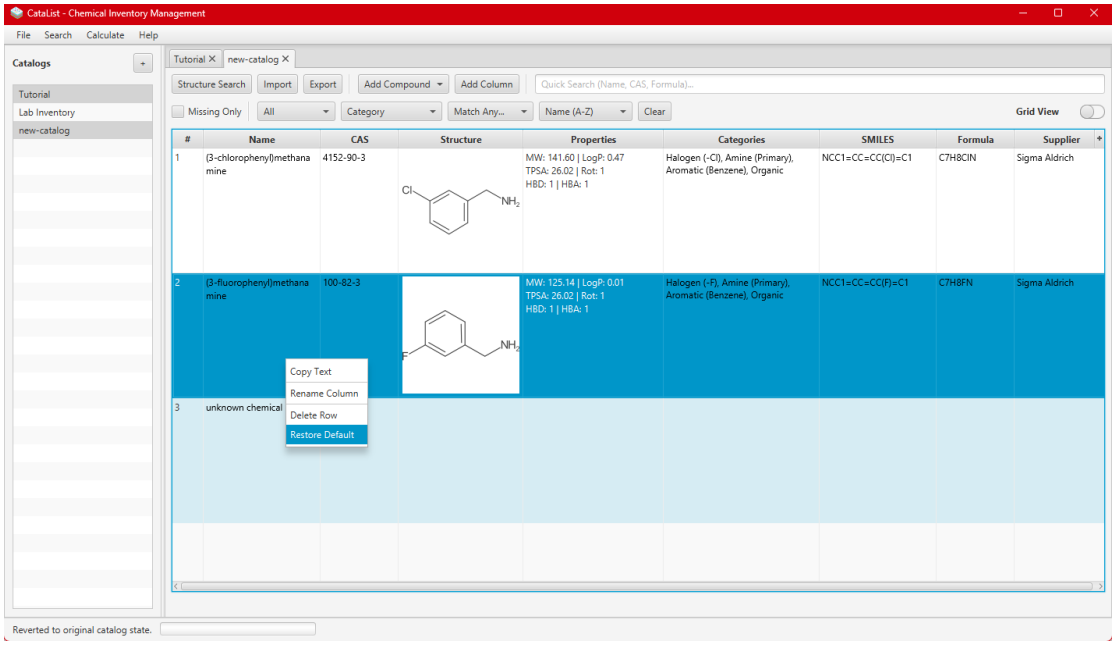
You may modify the Name, SMILES, or CAS fields of any entry.
After modification, **CataList** performs a new lookup based on the updated identifier and can overwrite existing data if a valid match is found.



Undo and Data Recovery
If a mistake is made:

- Press Ctrl + Z to undo the last action.
- Right-click an entry and select “Restore Default” to revert the row to its original imported state.

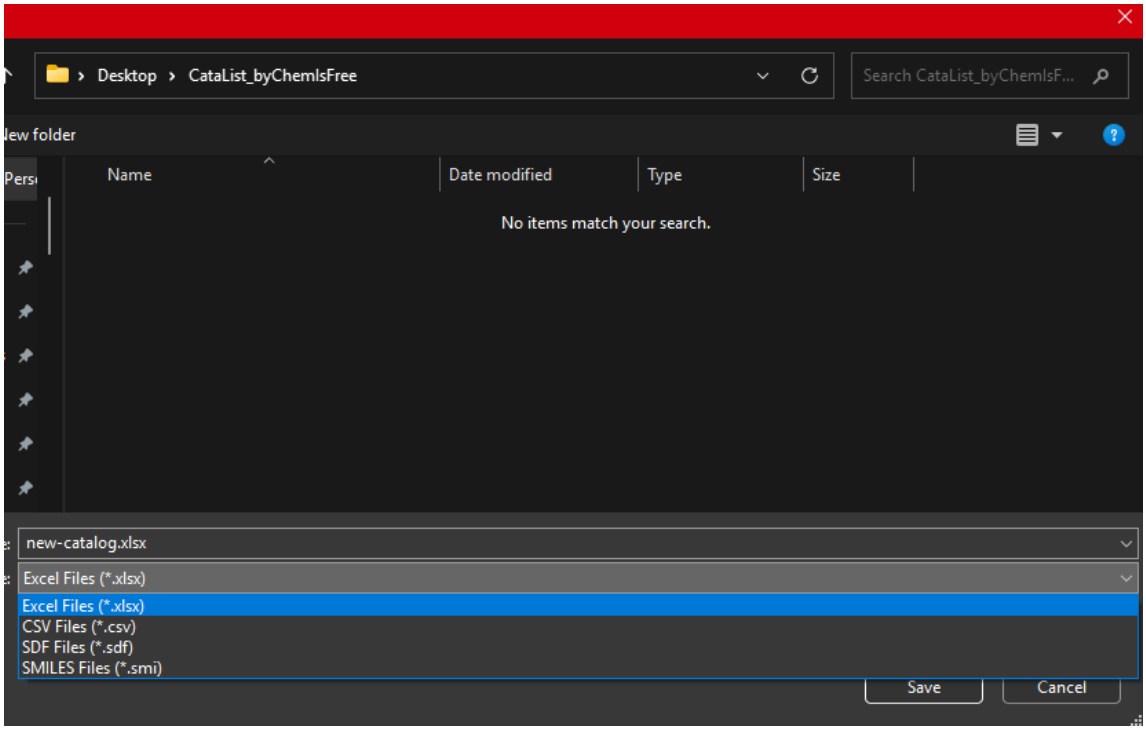
Handling Unresolved Compounds
Compounds with unresolved structures or missing SMILES are highlighted in blue for visibility.
By selecting the “Missing Only” checkbox, the catalog will display only unresolved entries. These can then be manually corrected or removed.



Saving and Exporting

CataList automatically saves catalogs before closing; however, maintaining a backup of the original file is strongly recommended. You can export your catalog at any time in .sdf, .csv, .xlsx, or .smi format.

This functionality also enables **CataList** to serve as a practical format-conversion tool for chemical libraries, allowing seamless interoperability with other cheminformatics software.



This marks the end of our tutorial. You are now ready to use **CataList**!

Some tips and best practices include:

- Naming the headers of your initial csv as “Name”, “CAS”, and “SMILES” for easier integration into the app. Even if you don’t, the app will either deduce it or ask you.
- Saving a copy of your initial database. Even though this app autosaves, and is robust, it never hurts to have a backup.
- Although rare, if the app crashes, restarting usually works.
- Donating and supporting our cause helps us create more and better tools. For more, visit therealchemisfree.com

*From the team of **ChemIsFree** and **CataList**, thank you!*